

EASWAR ANAND NARAYANAN

Postdoctoral Fellow | Southern Methodist University

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EDUCATION

Ph.D. in Physics, University of New Mexico, Albuquerque, USA Jan 2020 - May 2024

Thesis: *A Precise Measurement of the Lifetime of B_d^0 -mesons, Measurement of CP-violation Parameters of B_s^0 -mesons, the ATLAS Experiment at the LHC, and Development of Silicon Detectors for Future Particle Physics Experiments*

Advisor: Prof. Sally Seidel

M.S. in Physics, Indian Institute of Technology Madras, India Aug 2016 - July 2018

Thesis: *Sensitivity of Belle II experiment to the CP-violating parameter gamma*

Advisor: Prof. James Libby

B.S. with Honors in Physics, St. Stephen's College, University of Delhi, India Aug 2013 - May 2016

TECHNICAL SKILLS

Instrumentation, Device Physics & Semiconductor Technologies: Device physics, electrical characterization, semiconductor device testing, wafer-level testing, probe station operation, wafer-level defect analysis, semiconductor fabrication, microelectronics assembly and packaging, cleanroom protocols (ISO 7), statistical process control (SPC), design of experiments (DOE), yield analysis, quality assurance and reliability engineering, failure analysis, system-level stress validation, radiation hardness assurance, high-speed link characterization (BER testing, signal integrity, eye diagram analysis, SerDes characterization), precision measurement, bench testing, differential probing, hardware debugging, thermal modeling and mitigation strategies, CPU/GPU power and thermal optimization, optical metrology, digital microscopy, SEM, XRD, ESR, equipment calibration and maintenance, automated test equipment (ATE) operation

Embedded Systems, Hardware Integration & Communication Interfaces: Hardware–software interfacing, Python-based automation, USB and I2C communication protocols, ASIC communication and integration

Programming & Software Development: Python, C/C++, SQL, Bash, MATLAB, Mathematica, Git/GitHub, Jira, $\LaTeX$, Microsoft Office Suite

High-Performance & Distributed Computing: HPC cluster job scheduling, distributed computing, parallel processing, large-scale data workflows

Data Science, Statistical Analysis & Machine Learning: Data acquisition, preprocessing, feature engineering, statistical modeling, algorithm optimization, predictive modeling, simulation workflows, data reconstruction, data visualization; Libraries: NumPy, Pandas, SciPy, Matplotlib, SciKit-Learn, Minuit2, TensorFlow/Keras, PyTorch

Operating Systems: Linux, Windows, macOS

AWARDS & HONORS

Breakthrough Prize in Fundamental Physics, awarded as a member of the ATLAS Collaboration at CERN, recognized for contributions to experimental particle physics research 2025

Graduate Distinction, University of New Mexico 2024

Doctoral Conference Presentation Award sponsored by University of New Mexico 2023

Alexander Graham Bell Scholarship sponsored by Alexander Graham Bell Association for the Deaf and Hard of Hearing Graduate Students 2022

Chateaubriand Fellowship sponsored by the Embassy of France in the United States 2022-2023

Travel Award sponsored by Department of Particles and Fields, American Physical Society 2022

William G. Larsen, Ph.D., Memorial Award for the Best Teaching Assistant sponsored by University of New Mexico	2021
Sigma Xi Outstanding Graduate Researcher Award	2021
Junior Research Fellowship in Physics sponsored by University Grant Commission and University Lectureship in Physics for scoring in top 6 percentile in the National Eligibility Test	2018
Merit-Cum-Means scholarship sponsored by Indian Institute of Technology Madras for top 25% students	2017-2018
Institute Freeship Scholarship sponsored by Indian Institute of Technology Madras for top 10% students	2016-2017

RESEARCH & PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate, Southern Methodist University, Dallas, TX, USA June 2024 - Current
 Played a leading role in two parallel research efforts at SMU: hardware reliability and detector instrumentation R&D for the ATLAS Inner Tracker (ITk) upgrade supporting the High-Luminosity LHC program, and machine learning-based event classification for Higgs boson pair production (di-Higgs) searches using Run 3 ATLAS data.

- High-Volume ASIC Production Testing & Yield Recovery
 - Performed technical production qualification of 300+ GBCR ASICs (65 nm CMOS SerDes) within an 800-unit pre-production batch, including BER testing at 1.28 Gb/s, register validation, and parametric testing to meet IEEE 802.3 BER specifications ($< 10^{-12}$) for high-reliability data links. Established test methodology supporting large-scale volume production in collaboration with multi-site teams. Delivered qualified ASICs to CERN (April 2026) for optoboard assembly as part of the ATLAS ITk pixel detector production pipeline.
 - Assembled automated test platform using Xilinx KC705 FPGA with custom carrier boards, executing PRBS7 pattern generation, I2C register configuration at 100–400 kHz, and parallel 6-channel receiver characterization achieving 1 hour/unit throughput.
 - Performed root cause analysis on yield-limiting signal integrity failures using Tektronix DSA70804B oscilloscope and differential probing; analyzed eye diagrams to isolate faulty FPGA multi-gigabit transceivers (MGTS) and restore production yield.
 - Executed test coverage including functional BER validation, register scan, clock phase margin sweeps (16 phases), and equalization optimization across 256-point parameter space.
- Radiation Hardness Assurance & TID Reliability Testing
 - Developed failure analysis framework to assess GBCR ASIC reliability under Total Ionizing Dose (TID) exposure, characterizing radiation-induced soft errors (bit flips, logic faults) for long-term HL-LHC deployment validation.
 - Expanded TID characterization setup from single to 16 GBCRs using multiplexers and voltage level shifters, increasing parallel testing throughput by $16\times$.
 - Built Python-based real-time monitoring scripts logging 250,000+ data cycles across six months of continuous testing; automated GBCR configuration routines controlling six key registers (equalization, CDR, output drivers) for consistent operation across all devices.
 - Implemented USB and I2C communication at 100–400 kHz for stable parallel data acquisition; deployed experimental setup at Brookhaven National Laboratory for extended TID characterization studies.
 - Observed stable operation of the majority of ASICs after high-fluence irradiation, validating radiation tolerance for HL-LHC conditions.
- Efficiency Modeling & Statistical Calibration
 - Modeled joint trigger and offline b-tagging efficiency calibration in $bb\tau\tau$ Higgs final states using two conditional probability methods in Run 3 ATLAS data; validated efficiency extraction using $t\bar{t} \rightarrow e\mu$ events and performed likelihood fits across jet p_T and b-tagging score bins to derive trigger and offline scale factors with quantified uncertainties.

- Performed extended likelihood-based fits across 36 multidimensional feature bins to extract calibration factors correcting simulation mismodeling, maintaining efficiency above 80% in target regions with 1–4% uncertainty control.
- Multi-Modal GNN Development & Signal-Background Separation
 - Adapted and studied a multi-modal GNN classifier within the ATLAS Salt framework to distinguish ggF and VBF Higgs pair production signals from background in the hadronic $\tau\tau$ decay channel; achieving ROC AUC values above 0.98 compared to baseline BDT performance (~ 0.97).
 - Built with PyTorch Lightning, scaling model complexity from 2.3M to 9.6M parameters; accelerated inference by 35% through ONNX optimization and CUDA-based deployment.
 - Studied two architectures: a flat feature-based classifier and a multi-modal Transformer-based GNN with object-level modeling and cross-modality attention. The Transformer-based GNN demonstrates improved performance over BDT and flat classifiers in background rejection, score separation, and physics significance, achieving up to a 20% increase in background rejection at TPR > 0.75.
- Multi-Task Learning & Auxiliary Task Optimization
 - Conducted systematic ablation study on auxiliary task weighting in a multi-task Transformer model; discovered that light regularization (weight 0.001) outperformed data-driven class balancing by 20% in classification significance via 5-fold cross-validation. Investigated class weight optimization for extreme signal-background imbalance (614:1 ratios).
 - Processed 200+ GB of raw data into optimized HDF5 format, reducing storage footprint by 82%; executed large-scale training and inference on CERN's HPC infrastructure using distributed job scheduling, reducing model iteration time by 50%.

Visiting Research Fellow, IJCLab, University of Paris-Saclay, Orsay, France

Oct 2022 – July 2023

Contributed to the development and quality assurance of high-density silicon pixel modules for the ATLAS Inner Tracker (ITk) Phase-II upgrade at CERN, as part of a collaborative effort spanning 20+ institutions across Japan, Europe, and the U.S.

- Silicon Sensor Module Assembly, Testing & Quality Assurance
 - Engineered and tested 100+ high-density semiconductor pixel modules with 150 μm silicon sensors for next-generation ATLAS ITk detectors; conducted wafer-level electrical characterization using precision probe-station systems to validate sensor performance throughout the fabrication cycle.
 - Applied Design of Experiments (DOE) to accelerate inspection workflows, reducing optical metrology measurement time by 80% and defect scan times by 50% using advanced digital microscopy techniques.
 - Implemented Statistical Process Control (SPC) to monitor quality metrics for 100+ pixel modules in an ISO 7 cleanroom, identifying process deviations and upholding stringent environmental controls.
 - Automated data extraction workflows for optical metrology and probe-station instruments, enabling real-time visualization and analysis; built scripts to convert raw instrument outputs into database-ready metafiles, improving data traceability and integrity.
 - Automated QC pipelines and optimized production database integration, reducing bottlenecks and improving overall workflow efficiency across the production cycle.
 - Mentored junior staff in cleanroom assembly, testing protocols, and QA processes; authored technical documentation and progress reports supporting reproducibility and knowledge transfer across international detector development teams.

Graduate Research Assistant, University of New Mexico, Albuquerque, NM, USA

Jan 2020 – May 2024

Worked on multiple research projects within the ATLAS experiment at CERN, spanning precision B-physics measurements, silicon detector reliability modeling, trigger algorithm development, and detector operations.

- B-Physics Measurements (ATLAS Run 2)

- Developed trigger and event selection criteria from dominant triggers in ATLAS Run 2 data for both B_d^0 lifetime and B_s^0 CP-violation analyses; applied time-efficiency corrections to reduce proper decay time bias from trigger and selection cuts.
- Performed simultaneous mass and proper decay time likelihood fits to extract the B_d^0 meson lifetime in $B_d^0 \rightarrow J/\psi K^{*0}$ decays, improving statistical precision relative to previous measurements – contributing to one of the most precise B_d^0 lifetime measurements to date. Published in *Eur. Phys. J. C* 85 (2025) 736.
- Performed simultaneous likelihood fit for 9 physics parameters in $B_s^0 \rightarrow J/\psi \phi$ decays and $B_s^0 - \bar{B}_s^0$ mixing, achieving $1.32\times$ statistical precision improvement over the previous ATLAS result.
- Estimated systematic error from time efficiency correction on lifetime measurement, accounting for $\sim 37\%$ of total systematic error; developed methods to filter events with multiple and fake candidates.
- Radiation Damage Modeling in Silicon Sensors
 - Developed predictive models in Python and C++ to simulate ATLAS Pixel detector degradation under radiation and thermal stress using the Hamburg Model; improved simulation accuracy by 70% through real-world calibration and high-fluence data fitting (1.25×10^{15} 1 MeV N_{eq}/cm^2). Utilized Migrad and Hesse algorithms in Minuit2 for numerical minimization in a multivariate analysis.
 - Studied evolution of leakage current and depletion voltage for the ATLAS Pixel detector using temperature and fluence data from 2015–2023; evaluated radiation tolerance in Run 3 (2022–2025) and projected 10-year radiation impact on the ATLAS ITk Pixel for the HL-LHC phase.
 - Advocated scalable multithreaded simulation strategies as part of the Snowmass 2021 process; co-authored peer-reviewed publication on silicon radiation detector simulations for high energy physics.
- Irradiation Campaigns at LANSCE and FNAL
 - Applied the NIEL scaling hypothesis to compute target fluences, required proton counts, and irradiation durations for 200+ silicon sensor samples across beam campaigns at LANSCE and Fermilab, accounting for individual sample geometry and dosage requirements. Guided fellow group members on irradiation procedures based on these calculations.
- ATLAS High Level Triggers (HLT) and AthenaMT
 - Developed HLT algorithms for B-Physics within the ATLAS AthenaMT software framework; optimized for multithreaded CPUs and low memory usage.
 - Configured B-Physics triggers with transverse momentum threshold as low as 4 GeV; validated triggers at $>70\%$ efficiency using Monte Carlo samples.
- ATLAS Inner Detector – Run 3 Data-Taking Oct 2022 – April 2023
 - Carried out shift tasks in the ATLAS control room representing the University of New Mexico, focusing on the ATLAS Inner Detector during both shutdown and data-collection phases.

Graduate Teaching Assistant, University of New Mexico, Albuquerque, NM, USA

Jan 2020 – Dec 2023

Supported undergraduate teaching across multiple physics courses.

- Taught recitation classes, tutored and graded assignments for undergraduate courses in Classical Mechanics (PHYS 303/304), Quantum Mechanics (PHYS 491/492), Nuclear and Particle Physics (PHYS 450), and Geophysics (PHYS 327).

CONFERENCES & INVITED TALKS

B-Physics Results from ATLAS and CMS

Rencontres de Moriond 2025 - QCD and High Energy Interactions, La Thuile, Italy

April 2025

Radiation Damage Effects in ATLAS Pixel Detector Monte Carlo Simulations: Status and Perspectives

Technology and Instrumentation in Particle Physics (TIPP) 2023, Cape Town, South Africa

Sept 2023

Measurement of the CP-violating phase ϕ_s in $B_s^0 \rightarrow J/\psi\phi$ decays in ATLAS/LHC at 13 TeV LHCC Meeting, Geneva, Switzerland	Nov 2022
Mixing and CP- Violation at the LHC Vietnam Flavour Physics Conference, Quy Nhon, Vietnam	Aug 2022
Measurement of the CP-violating phase ϕ_s in $B_s^0 \rightarrow J/\psi\phi$ decays in ATLAS/LHC at 13 TeV APS April Meeting, New York City, USA	April 2022
Measurement of the CP-violating phase ϕ_s in $B_s^0 \rightarrow J/\psi\phi$ decays in ATLAS/LHC at 13 TeV Meeting of the Division of Particles and Fields of the American Physical Society	July 2021
New Techniques for Finding New Physics at the LHC University of New Mexico Seminar	April 2021
Measurement of the CP-violating phase ϕ_s in $B_s^0 \rightarrow J/\psi\phi$ decays in ATLAS/LHC at 13 TeV APS Four Corners Section Annual Meeting	Oct 2020

SELECTED PUBLICATIONS

Improved analysis of non-resonant Higgs boson pair production in the $b\bar{b}\tau^+\tau^-$ final state with 196 fb⁻¹ of data collected at $\sqrt{s} = 13$ TeV and 13.6 TeV with the ATLAS detector. ATLAS Collaboration, Submitted to EPJC, 2026

Precision measurement of B^0 meson lifetime in $B_d^0 \rightarrow J/\psi K^{*0}$ decays with the ATLAS detector. ATLAS Collaboration, Eur. Phys. J. C 85 (2025) 736, 2025

Simulations of Silicon Radiation Detectors for High Energy Physics. B. Nachman, et al, APDL-2022-002 (2022), 2022

Co-author on 470+ peer-reviewed collaboration publications with the ATLAS Collaboration at CERN; complete publication record and author-list contributions available via Inspire-HEP.

PROFESSIONAL SERVICE

Peer Reviewer, Journal of Instrumentation (JINST), IOP Publishing 2025-2026

Reviewed manuscripts on radiation damage and electrical characterization of silicon-based semiconductor detectors for high energy physics applications.

OUTREACH

Poster Judge, Undergraduate Research Opportunity Conference (UROC), University of New Mexico 2022, 2024

General Secretary to the Enabling Unit, at St. Stephen's College, Delhi, an university initiative for supporting disabled students 2016

Organizing Committee, Inter-college Fest for Disabled Students, St. Stephen's College, Delhi 2015

Facilitator, Science Facilitation Centre, Mar Ivanios College, University of Kerala 2015